Evaluating Credit Risk Through Transaction-Level Cash Flows: A Data-Driven Alternative to Traditional Scoring

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Abstract

Traditional credit scores remain standard to credit risk evaluation, yet their heavy reliance on historical repayment data systematically excludes borrowers with limited or no formal credit history. This dependence not only reinforces structural inequities but also fails to incorporate the rich behavioral signals embedded in individuals' day-to-day financial activity. As access to digital banking expands, large-scale transaction data provide an opportunity to rethink how creditworthiness is assessed.

This project develops a cash-flow underwriting framework that leverages income consistency, spending composition, liquidity trends, and short-term financial volatility to predict default risk. By transforming raw bank account and transaction records into structured, consumer-level behavioral features, we construct models that capture real-time financial stability rather than past borrowing outcomes alone. Our results demonstrate that behavior-based cash-flow signals can meaningfully enhance risk prediction while supporting a more inclusive and data-driven approach to credit evaluation.

Website: https://harisdsc.github.io/DSC180B
Code: https://github.com/harisdsc/DSC180B

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1 Introduction

Credit scores play a critical role in shaping an individual's financial opportunities, influencing access to loans, credit cards, and housing. However, traditional credit scoring systems are inherently biased toward individuals who already have established credit histories. This creates a structural disadvantage for the credit-invisible group, including young adults, recent immigrants, or individuals from low-income backgrounds, who may have limited or no prior credit activity. As a result, credit scores often reflect past debt repayment behaviors rather than a borrower's true real-time financial reliability.

Let $X_i = \{x_{i1}, x_{i2}, \ldots, x_{iT}\}$ denote the sequence of financial transactions for the individual i where T represents the number of transactions of that individual, where each transaction x_{it} includes features such as the amount, merchant category, and the date of transaction. Let $y_i \in \{0, 1\}$ indicate the observed credit outcome (e.g., non-delinquent or delinquent) for that individual. Traditional credit scoring models compute y_i as a function of historical credit activity C_i:

$$\hat{y}_i^{\text{trad}} = f_{\text{credit_score}}(C_i)$$

where C_i is undefined for individuals with a limited credit history. This induces structural bias where the chances of being classified as delinquent is higher for the credit-invisible group.

$$P(\hat{y}_i^{\text{trad}} = 1 \mid \text{no prior credit}) > P(\hat{y}_i^{\text{trad}} = 1 \mid \text{rich credit history})$$

Our objective is to define a cash-flow-based predictive function:

$$\hat{y}_i^{\text{cash-flow}} = g(X_i)$$

that estimates creditworthiness using bank transaction data alone.

This project seeks to address the current limitation by leveraging transactional financial data to predict credit risk without relying on pre-existing credit scores. By analyzing consumer bank transaction histories, including spending patterns, income regularity, and cash flow volatility, this project explores how machine learning and large language models can provide a fairer, data-driven alternative to traditional credit scoring. This approach, known as cash-flow underwriting, evaluates creditworthiness based on observed income and expenditure behavior rather than repayment history, thereby extending access to those excluded by legacy scoring systems.

1.1 Previous Work

Fueled by advances in data aggregation, open banking, and the rise of digital financial infrastructure, cash-flow underwriting has gained significant traction in the past decade. Fintech innovators such as Prism Data have introduced proprietary frameworks like CashScore,

which assess credit risk exclusively using cash-flow information. According to Prism Data, "CashScore provides an average of 30% predictive lift above traditional credit scores alone," indicating that combining cash-flow insights with conventional scores can substantially improve credit risk prediction (PrismData 2025). Similarly, a 2025 study by the National Bureau of Economic Research (NBER) found that lenders using business-checking account cash flows achieved higher loan approval rates for younger entrepreneurs compared with those relying solely on credit scores (Hair et al. 2025). Building on this momentum, our project develops a customized implementation of cash-flow underwriting aimed at giving financially excluded consumers a second opportunity to demonstrate creditworthiness.

1.2 Data

For this project, we utilized four datasets provided by Prism Data and sourced from participating financial institutions. These datasets contain detailed account and transaction information used for model development and evaluation. Importantly, consumers retain control over which and how many bank accounts they elect to share for underwriting purposes.

Below, we describe each dataset used in the model training process.

Table 1: Consumer-Level Dataset (consDF)

consDF (15000, 4)		
dataframe containing consumer-level information with a holdout set of 3000 null `DQ_TARGET`		
prism_consumer_id	str	Unique ids for each Prism Data consumer
evaluation_date	datetime	The date of evaluation
credit_score	float	Credit score of the consumer
DQ_TARGET	float	Binary outcome determining delinquency

Table 2: Account-Level Dataset (acctDF)

acctDF (24466, 5)		
dataframe containing account-level information		
prism_consumer_id	str	Unique ids for each Prism Data consumer
prism_account_id	str	Unique ids for each Prism Data consumer account
account_type	str	Type of account
balance_date	str	Date of balance
balance	float	Balance amount

Table 3: Transaction-Level Dataset (trxnDF)

trxnDF (6407321, 6)		
dataframe containing transaction-level information with transaction categories encoded		
prism_consumer_id	str	Unique ids for each Prism Data consumer
prism_transaction_id	str	Unique ids for each consumer transaction
category	int	Category of transaction
amount	float	Amount of transaction
credit_or_debit	str	Inflow or outflow transaction
posted_date	str	Date of transaction

Table 4: Category Mapping Dataset (cat_map)

cat_map (50, 2)		
a mapping from integer category codes to category names. Categories 5, 10, and 15 are deprecated		
category_id	int	Integer category codes
category	str	Transaction category names

1.2.1 Data Privacy

The datasets used in this project are de-identified prior to analysis. Personally identifiable information, such as names and account numbers, are not included in the modeling dataset. Instead, consumers are represented by anonymized unique Prism Consumer IDs, Prism Account IDs, and Prism Transaction Ids instead.

All analysis focuses on behavioral patterns (e.g., income stability, spending ratios, liquidity trends) rather than individual identity. This ensures that the model evaluates actual financial behavior rather than demographic or personally sensitive attributes.

In addition, access to the data is restricted to authorized project members and used exclusively for academic research purposes.

1.2.2 Data Limitations

While transaction-level bank data provides a rich and dynamic view of consumer financial behavior, several limitations should be noted.

First, because consumers control which accounts are shared, the dataset may not capture a complete view of an individual's financial situation. For example, certain income sources, liabilities, or external obligations may be unobserved. This partial observability can introduce measurement error in derived features such as spend-to-income ratios or net cash flow.

Second, transaction categorization may contain noise or inconsistencies depending on how

merchants are labeled. Although we rely on provided category mappings, prior misclassification can affect category-level feature construction.

Third, the historical depth of available data varies across sources of providers causing some individuals to have multiple years of transaction history, while others may have only a few months. To mitigate this issue, we incorporate rolling-window features and track the number of observed months per consumer to contextualize model outputs.

Despite these limitations, transaction-level cash-flow data offers meaningful behavioral insights that traditional credit bureau data may not capture, particularly for consumers with limited credit history.

2 Methods

2.1 Data Cleaning

To ensure model integrity and prevent data leakage, we implemented the following data screening and exclusion rules:

2.1.1 Screening

1. **Holdout of Unlabeled Consumers.** We separately stored approximately 3,000 consumers with null `DQ_TARGET` values for later out-of-sample evaluation. These consumers were excluded from model training to avoid introducing label ambiguity.
2. **Removal of Credit Scores.** The `credit_score` variable was dropped from `consDF`. Our objective of this project is to evaluate the predictive power of transaction-based cash-flow features. Including credit scores would bias results and undermine the independence of the proposed Cash Score. This variable was provided for later observations on how our cash scores compare to traditional scoring metrics.
3. **Separation of Evaluation Dates from Features.** Evaluation dates were excluded from the training feature set, as they are administrative markers rather than behavioral predictors.
4. **Filtering of Transactions.** We removed all rows from `trxnDF` where `posted_date` occurred after the consumer's evaluation date recorded in `consDF`. Since delinquency is assessed at the evaluation date, transactions occurring afterward must not influence prediction. This step prevents forward-looking bias (data leakage).
5. **Duplicate Removal.** Duplicate transaction records in `trxnDF` were removed to ensure accurate aggregation and feature construction.
6. **Date Standardization.** The `posted_date` field in `trxnDF` was converted from string format to a datetime object to enable proper temporal filtering, rolling window calculations, and chronological sorting.

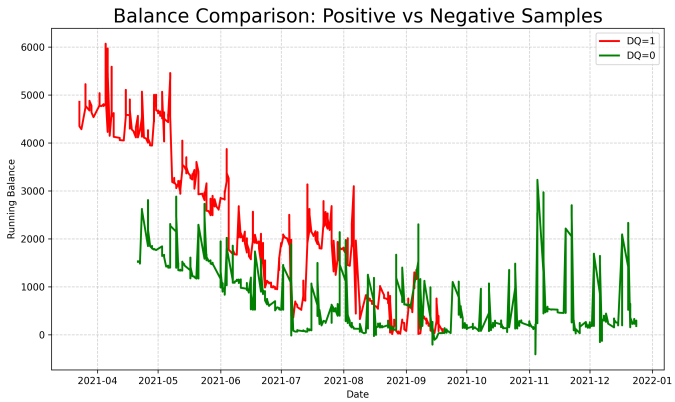
2.1.2 Exclusion Rules

To ensure reliability and interpretability of the Cash Score, we apply minimum data sufficiency criteria.

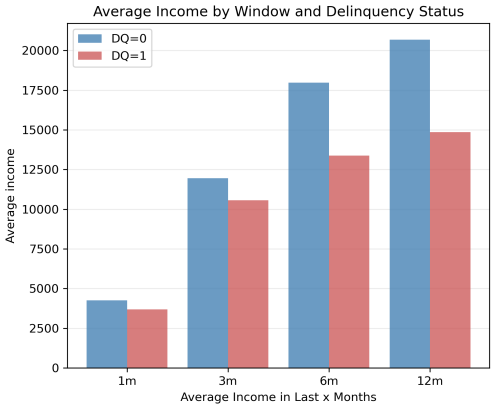
1. **No Linked Accounts** Consumers must link at least one active bank account for scoring. Our Cash Score relies on transaction-level cash-flow data sourcing from bank accounts. The absence of linked accounts leads to the absence of transaction data, preventing the construction of behavioral features such as income inflows, spending patterns, and liquidity trends.
2. **Minimum Transaction History** Consumers must have at least 3 months of transactions prior to their evaluation date to receive a Cash Score. Insufficient transaction history limits the model's ability to capture income stability, spending behavior, and liquidity trends.

Consumers violating these rules will temporarily be ineligible for scoring, as the model cannot generate a meaningful risk assessment with the current data provided. This exclusion is based solely on data adequacy and does not reflect any demographic or credit-based screening. As a result of these cleaning procedures and exclusion rules, 2681 consumers were removed from the training datasets.

2.2 Exploratory Data Analysis (EDA)



(a) Running Balance Trajectories of DQ=0 & DQ=1 Consumers



(b) Average Income of DQ=0 & DQ=1 Consumers in the Past x Months

Figure 1: Data Patterns Observed During Exploratory Data Analysis

Before creating features, we performed exploratory data analysis to identify patterns that could help differentiate between delinquent and non-delinquent consumers. The dataset exhibits a notable class imbalance, with approximately 10% of consumers labeled as delinquent (DQ = 1) and roughly 90% labeled as non-delinquent (DQ = 0).

Further analysis revealed behavioral differences between these two groups. As shown in Figure 1a, delinquent consumers tend to exhibit higher balance volatility and more pro-

nounced downward trends over time, suggesting greater financial instability and susceptibility to liquidity shocks. In contrast, non-delinquent consumers generally maintain more stable balances with fewer extreme fluctuations.

Figure 1b compares the average income levels of delinquent and non-delinquent consumers across multiple historical windows (1, 3, 6, and 12 months). Across all windows, non-delinquent consumers consistently show higher average income levels than delinquent consumers, and the gap widens as the time horizon increases. This pattern suggests that sustained income stability over longer periods may be associated with lower delinquency risk.

These exploratory findings highlight meaningful differences in liquidity stability and income dynamics between the two groups, motivating the inclusion of features related to balance behavior, income magnitude, and income consistency in the subsequent modeling framework.

2.3 Features

While the provided datasets include detailed transactional and account-level information, additional features must be engineered to transform these records into structured behavioral indicators that are more directly informative for repayment risk prediction. Here is a general overview of our engineered features:

- **Income Features** focus on the magnitude, stability, and consistency of inflows. We aggregate income across defined income-related transaction categories and derive metrics such as total and average monthly income, income volatility, number of income sources, income regularity across months, and recency of income events.
- **Category Spending to Income Ratio Features** capture how consumers allocate income across specific expense categories over time. By evaluating these ratios across multiple temporal windows (e.g., 1, 3, 6, and 9-month periods), we are able to summarize how consumers allocate income toward specific expense categories over time. These summaries capture a consumer's typical spending intensity relative to income and reflect structural consumption burdens that may influence repayment capacity. These features are particularly informative for assessing financial discipline, budget stability, and potential liquidity stress, all of which are relevant indicators of repayment capacity.
- **Balance Dynamic Features** leverage reconstructed running balances to characterize temporal financial behavior. These include balance volatility, daily changes, drawdowns, recent balance trends (e.g., 30, 90, 365 days), transaction frequency, credit–debit flow patterns, balance trends, and extreme event flags such as overdrafts or unusually large transactions. Together, these features capture both short-term shocks and longer-term financial trajectories.
  To accurately track consumer-level financial dynamics over time, we constructed a running balance dataframe that reconstructs each consumer's balance history using transaction-level data. We began by computing an initial balance snapshot for each consumer, defined as the earliest available balance date, and used this as a reference point. Using the initial snapshot, balances were propagated forward in time by

cumulatively summing signed transactions occurring after the snapshot date, and backward in time by reversing the cumulative sum for transactions that occurred prior to the snapshot.
Constructing a running balance is critical because it transforms raw transaction logs into a coherent time series representation of financial behavior. This allows us to capture liquidity dynamics such as cash flow volatility, balance trends, and to identify temporal patterns.

- **Account-Level Features** summarize overall financial structure, including total and average balances, account diversity, liquidity indicators, cross-account balance dispersion, negative balance exposure, and wealth tier flags. These features provide a snapshot of consumers' available resources and account composition.
- **Temporal Behavior Features** capture patterns in transaction timing to identify consumer financial habits and stability. Because most consumer histories span less than 180 days, we avoid coarse seasonal indicators that could leak dataset structure into the model. Instead, we focus on high-frequency temporal patterns. In the time domain, we detect recurring habits such as spending on specific weekdays or bill payments on fixed days of the month. In the frequency domain, we apply Continuous Wavelet Transforms (CWT) to convert transaction histories into spectral signals, revealing the regularity of financial activity. Strong periodic signals indicate stable income or bill cycles, while irregular patterns suggest income volatility or unstable spending behavior.

All engineered features were ultimately aggregated to the consumer level for model training, ensuring that each row in the final dataset represents a single consumer.

2.3.1 Feature Selection

To improve model interpretability and prevent overfitting, we applied a multi-stage feature selection process that systematically removed redundant, uninformative, or unstable predictors. First, we applied L1-regularized (Lasso) logistic regression, which introduces an absolute value penalty on feature coefficients and drives the weights of less predictive variables to zero, effectively performing embedded feature selection. Next, we eliminated features with zero variance, as variables that do not vary across observations provide no predictive information. We then conducted collinearity screening by computing pairwise correlations and removing features with correlation coefficients greater than 0.85 to reduce redundancy and improve model stability. Finally, we performed manual inspection of the remaining features to remove variables that were difficult to interpret or unlikely to generalize, ensuring that the final feature set retained strong predictive power while remaining transparent and interpretable for downstream analysis.

2.4 Model

We trained the following models with a 80-20 train-test split on the consumer level and performed hypertuning using Optuna.

Logistic Regression
We used Logistic Regression as our baseline model because it could capture the basic linear relationships our features have with `DQ_TARGET`. Features are standardized with `Standard Scalar`, and the model is trained with class balance to account for the class imbalance in the dataset. Model performance is evaluated using the AUC-ROC score on both the train and test sets to test for potential overfitting.

LightGBM
LightGBM is a high-performance gradient boosting framework that uses tree-based learning algorithms. LGBM grows trees leaf-wise rather than level-wise, which often leads to faster convergence and improved accuracy, especially for large datasets. LGBM also supports handling class imbalance through the `sample_weight` parameter, which is critical given the skewed target distribution in our dataset. However, like other gradient boosting methods, LGBM can overfit if hyperparameters are not carefully tuned, so we included parameters like `reg_alpha` and `reg_lambda` to regularize the data.

XGBoost
To capture more complex, non-linear relationships in the model, we elected to use XGBoost, which is a high-performance implementation of gradient boosted decision trees. It builds an ensemble of decision trees sequentially, which allows for it to learn non-linear feature interactions. We conducted grid search cross-validation to look for the best hyperparameters and used the AUC-ROC score as an evaluation metric. XGBoost also supports class imbalance through `scale_pos_weight`, which is critical as our dataset has a skewed target distribution. The performance of the model is evaluated using the AUC-ROC score, and latency per sample is measured to ensure deployment feasibility. XGBoost also includes built-in regularization (reg_alpha and reg_lambda) to prevent overfitting.

CatBoost
To further capture complex, non-linear relationships in the data, we also included CatBoost, a gradient boosting model that is particularly effective when working with structured tabular datasets. CatBoost builds an ensemble of decision trees sequentially, similar to other boosting methods, but is designed to reduce prediction shift and improve generalization through its ordered boosting procedure. It can also handle categorical information efficiently and is robust to overfitting when properly tuned. We performed hyperparameter tuning using Optuna to identify the best-performing settings and evaluated model performance using the AUC-ROC score on both the train and test sets. Because our dataset has a skewed target distribution, CatBoost was trained with class weighting to account for class imbalance.

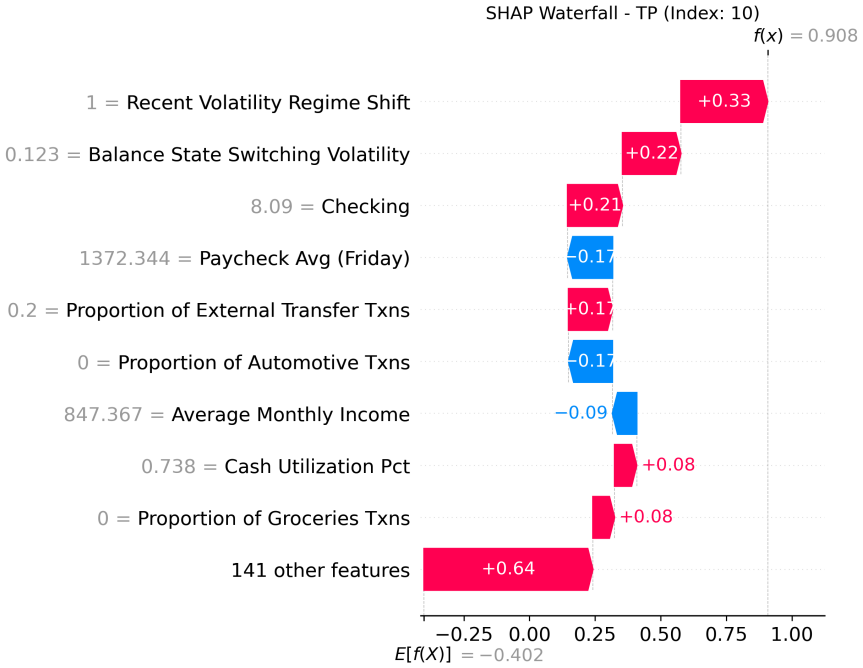


Figure 2: SHAP Waterfall Plot for a Single Consumer

2.5 Reason Codes

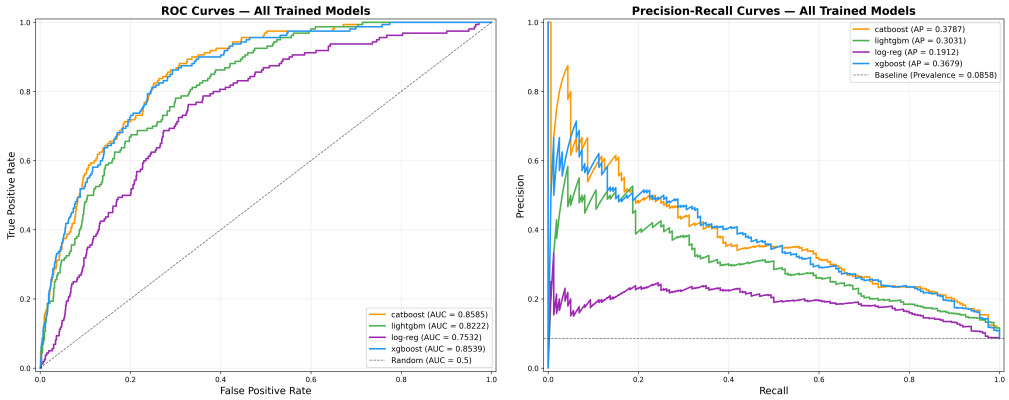
After training and evaluating multiple models, we selected our best-performing model to generate reason codes explaining why a consumer receives a particular score. To comply with the Fair Credit Reporting Act (FCRA), our model reports the top three factors that most negatively influence an applicant's score. These explanations are generated using SHAP (SHapley Additive exPlanations), a model interpretation method that quantifies the contribution of each feature to an individual prediction. For every consumer evaluated by the model, SHAP values measure how much each feature increases or decreases the predicted probability of delinquency relative to the model's baseline output. These contributions can be visualized using a waterfall-style explanation that illustrates how behavioral signals derived from a consumer's transaction history—such as liquidity levels, spending intensity, or balance trends—shift the predicted risk score away from the population baseline.

The features with the largest positive contributions to the consumer's approval score are identified as the primary drivers of risk. To convert these technical feature contributions into human-readable explanations, each feature is mapped through an internal `FCRA_MAPPING` dictionary that translates model variables into standardized explanation templates. This mapping step automatically produces concise English reason codes, such as "Average Monthly Income" or "Proportion of Grocery Transactions," which can be included in adverse action notices. This process ensures that each model decision can be traced to specific behavioral signals while producing explanations that are interpretable for consumers and compliant with FCRA requirements.

In Figure 2, the prediction begins at the model's average output $E[f(X)] = -0.402$ and is ad-

justed step-by-step (bottom-up) by each feature's SHAP value, either increasing or decreasing the predicted risk. Red bars indicate features that increase the predicted delinquency risk, while blue bars represent features that decrease it. In this example, the strongest contributors to higher risk include a recent volatility regime shift (+0.33), high balance state switching volatility (+0.22), and the consumer's checking account balance (+0.21), suggesting unstable liquidity patterns and irregular balance behavior. Additional risk is associated with a higher proportion of external transfer transactions and higher cash utilization. In contrast, several factors mitigate risk: higher paycheck averages, the absence of automotive-related transactions, and higher average monthly income all reduce the predicted probability of delinquency. The combined effects of these features shift the prediction from the baseline to the final model output $f(x)$, which is close to 0.9, indicating a high predicted likelihood of delinquency for this consumer.

3 Results



(a) AUC-ROC (b) Precision-Recall

Figure 3: Performance Metric Comparison Across All Models

Table 5: Model Performance + Latency Comparison

model	AUC-ROC	inference latency
log-reg	0.7532	0.4494 ms
lightgbm	0.8222	0.5535 ms
xgboost	0.8539	1.3380 ms
catboost	0.8585	0.7115 ms

From the Figure 3 and Table 5 above, we observe that the best performing model is the CatBoost model we implemented, achieving a AUC-ROC score of 0.8585 on the test set.

All models produced a probability of default using the `predict_proba` classifier method.

This probability is mapped for better interpretability using the formula,

$$10 * (1 - \hat{P}_{\text{default}})$$

producing a Cash Score, rounded to two decimals, on a 0–10 scale that is independent of consumer credit scores. Similar to conventional scoring systems, higher scores indicate lower delinquency risk.

4 Model Outputs

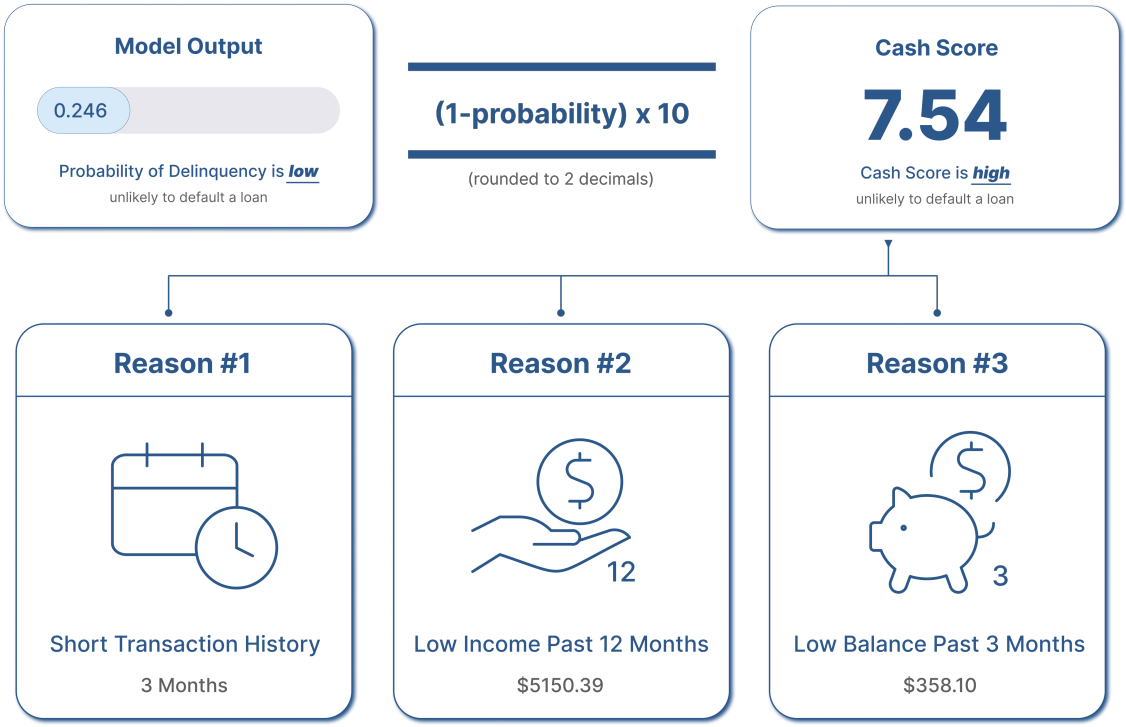


Figure 4: Model Outputs

The model produces two primary outputs for each consumer: a Cash Score as well as the top 3 primary factors influencing the model's prediction for each consumer. These reason codes highlight key behavioral signals that contributed to the risk assessment. Providing reason codes is important for transparency and helps ensure compliance with the Fair Credit Reporting Act (FCRA), as mentioned before, which requires that consumers be given understandable explanations for adverse credit-related decisions.

5 Discussion

When selecting the best-performing model, we focused on the AUC–ROC performance metric rather than precision and recall. In our setting, the model is designed to produce a con-

tinuous risk score that can be translated into different approval thresholds depending on a lender's risk tolerance. AUC–ROC evaluates the model's ability to correctly rank consumers by risk across all possible classification thresholds, rather than measuring performance at a single fixed cutoff as precision and recall do. This property makes AUC–ROC particularly suitable for real-world lending environments, where different lenders may adopt different approval thresholds based on their portfolio strategies and risk appetites. Higher thresholds reduce expected default risk but also decrease lending coverage. If a single global threshold were imposed, consumers whose scores fall below that cutoff could effectively be rejected by all lenders, limiting flexibility in credit allocation. By focusing on AUC–ROC, we evaluate the model's overall discriminatory power—its ability to separate delinquent from non-delinquent consumers—while preserving flexibility for lenders to determine their own operational thresholds.

6 Conclusion

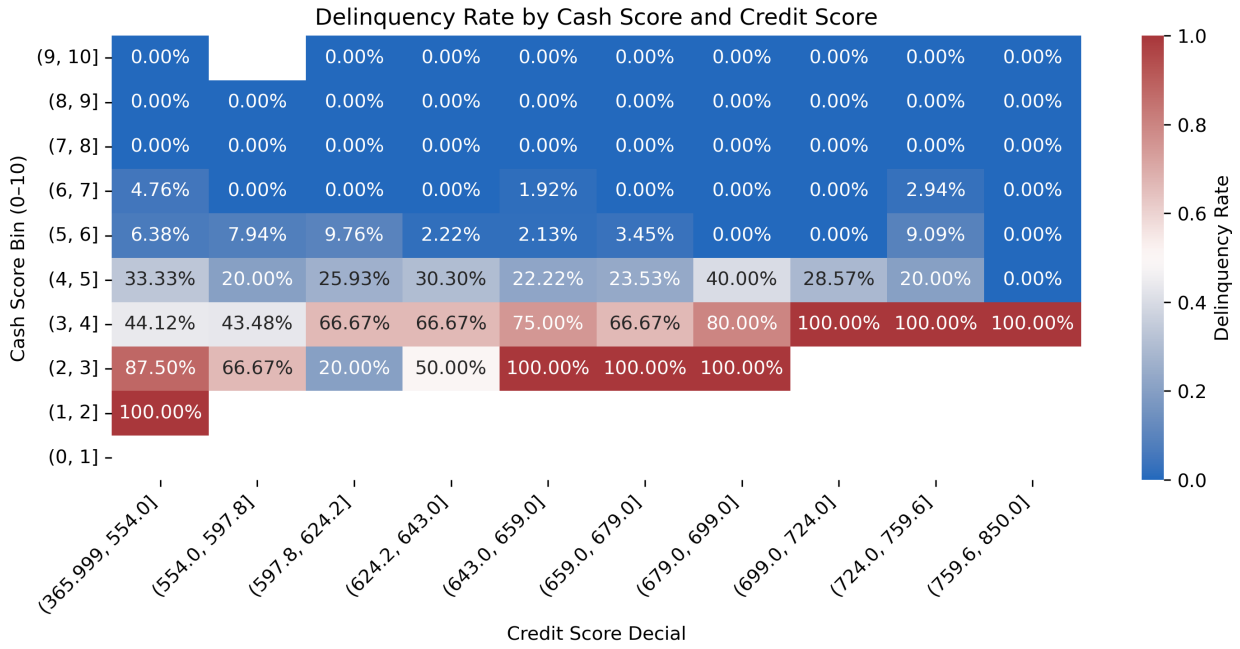


Figure 5: Delinquency Rates of Cash Scores Within Credit Bins

Figure 5 shows the relationship between the Cash Score and delinquency outcomes across different credit score quantiles. Each cell represents a group of consumers defined by a Cash Score bin (rows) and a credit score quantile (columns). The color of each cell indicates the delinquency rate within that credit group, where darker red corresponds to higher delinquency rates and darker blue corresponds to lower delinquency rates. The percentage displayed in each cell represents the proportion of consumers in that specific group who became delinquent. Importantly, the color scale is normalized within each credit score bin, allowing the visualization to highlight how delinquency risk varies across Cash Score

levels while holding credit score ranges constant. For example, within the credit score quantile (597.8, 624.2], consumers with Cash Scores between (2,3] show delinquency rates near 20%, while consumers with Cash Scores above (8,9] show near 0% delinquency. This pattern illustrates that even among borrowers with similar credit scores, the Cash Score provides additional differentiation in repayment risk.

As you can see, Cash Scores are not intended to replace traditional credit scores, but rather to support them. While credit scores often rely on fixed cutoff thresholds for approval decisions, they may not fully capture short-term liquidity conditions or behavioral risk differences among borrowers with similar scores. This is where our scores jump in.

By providing additional screening within each credit score band, the Cash Score helps differentiate repayment risk among consumers who appear identical under traditional scoring models. For example, individuals with high credit scores but weak cash-flow stability may still face elevated default risk. Incorporating the Cash Score therefore enhances risk stratification and supports more informed lending decisions.

Glossary

Cash Score - the score assigned to a consumer's credit evaluation produced by our Cashflow-Underwriting model. This score is our implementation of Prism Data's CashScore®. 6

Credit Invisible - the population of individuals who lack a history of borrowing/repaying money, therefore making them unable to be scored or approved for conventional loans. This group often faces difficulty accessing financial products like housing, auto loans, and credit cards. 3

Data Leakage - occurs when a model is trained using information that would not be available at the time a prediction is made. This often leads to artificially high performance because the model indirectly incorporates future outcomes. 6

Delinquent - refers to a debt (e.g. loan, credit card balance) that has not been paid by the scheduled due date. 3

Fair Credit Reporting Act - a U.S. federal law that regulates how consumer credit information is collected, used, and shared by credit reporting agencies and lenders. It is designed to protect consumer privacy, ensure the accuracy of credit reports, and provide transparency in lending decisions. Under the FCRA, lenders must provide adverse action notices when a consumer is denied credit or receives less favorable loan terms based on information in their credit report. 11

Optuna - an automatic hyperparameter optimization library for Python that helps find the best parameters for a machine learning model without manually trying combinations. 9

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